

VASSANTH MATHAN

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EDUCATION

M.S., Bioinformatics

Boston University

B.S., Microbiology, minor in Chemistry

New Mexico State University

SKILLS

Computational: HPC, R (including Shiny), Python, Unix, GitHub, Docker, SQL, Nextflow, Snakemake, ML, CLC Genomic Workbench, Geneious Prime, MATLAB, plink, gnomAD, Cytoscape.

NGS Techniques: Single cell, spatial, ATAC, RNA, ChIP, and Ribo-seq.

Microbiology: Fluorescence microscopy, mass spectrometry, qPCR, autoclaving, aseptic techniques, culturing and microscopy techniques, differential tests, Western Blot, ELISA, LFT.

PROFESSIONAL EXPERIENCE

Research Assistant – Thiagarajah-Silvester Lab

December 2025 - Present

Boston Children's Hospital

PIs: Jay Thiagarajah MD, PhD & Jocelyn Silvester MD, PhD

- Built and executed end-to-end NGS analysis pipelines (deconvolution, pre-processing, quality control, sc-RNAseq, sc-ATACseq, bulk-RNAseq and postprocessing workflows) using nextflow and verion control using Git to analyze genetic and cellular mechanisms of gastrointestinal tract malignancies.
- Troubleshoot technical issues in 10x Genomics workflows, including NGS pre-processing and QC issues, enabling successful data recovery from initially faulty runs.
- Managed large-scale genomic data storage, transfer and computational workflows in the HPC and AWS infrastructure.
- Collaborate on projects spanning NGS and clinical data analyses on Congenital Diarrheas and Enteropathies, celiac disease, and gastrointestinal tract malignancies.
- Contribute to research presentations, reports and publications.

Graduate research Intern – Thiagarajah-Silvester Lab

July 2025 – December 2025

Boston Children's Hospital

PIs: Jay Thiagarajah MD, PhD & Jocelyn Silvester MD, PhD

- Performed trajectory analyses using CellRank 2 to characterize intestinal epithelial differentiation, identifying late enterocyte subtype bifurcation patterns; presented computational findings to wet-lab audiences.
- Built a replicable bulk-RNA sequencing analysis pipeline and analyzed the GUT-AGE samples.
- Developed and optimized bioinformatic workflows to analyze molecular and cellular mechanisms of gastrointestinal tract malignancies.
- Performed biostatistical analysis and data quality assessments to ensure clinically robust and reliable results.
- Collaborate on multiple ongoing projects spanning computational biology, clinical data analysis, and translational research initiatives
- Contributed to research presentations and technical reports.

Volunteer Research Assistant – Segre Lab (Systems Biology)

September 2024 - May 2025

Boston University

PIs: Daniel Segre PhD & Ilija Dukovski PhD

- Optimized workflows in COMETS (Computation of Metabolic models in Time and Space) software to enhance predictive accuracy, integrated systems biology principles, flux omics and the COBRA (Constraint-Based Reconstruction and Analysis) methodology.
- Pioneered the use and development of COMETS to expand its application scope, validate accurate prediction for models to potentially enable stoichiometric analysis of tumor models linking

metabolism with tumor heterogeneity.

Volunteer Research Assistant – Systems Genomics and Genetics Lab
Boston University

September – December 2024
PI: Lei Hou, PhD

- Curated and analyzed over 30 microglial sequencing datasets from RPE, retinal, and brain cells, to uncover genetic variant impacts on TEG (TF-Enhancer-Gene) networks.
- Investigated transcription factor (TF)-mediated regulation in neurological diseases, understanding gene expression mechanisms for disease-driving factors.
- Performed genomic analysis workflows using TOBIAS (TF Occupancy prediction By Investigation of ATAC-seq Signals), QC and analysis of ATAC-seq, ChIP-seq, and scRNA-seq data.

Undergraduate Research Assistant - Metagenomics Lab
New Mexico State University

Aug 2023 - May 2024
PI: Jiannong Xu, PhD

- Analyzed the gut microbiome of diverse mosquito species by building a bioinformatics pipeline from scratch.
- Findings included novel species identification and geographic correlation inference with mosquito borne diseases.
- Performed functional annotation in MAGs, de novo assembly, RNA-seq (by transcript assembly), gene prediction, phylogenetic analysis and these manipulation works were done utilizing CLC genomic workbench, HPC, Python (Bowtie, MetaPhlAn, MicrobeAnnotator), MEGA11 and MEGAN6.
- Presented poster: A bioinformatics pipeline for characterizing microbiome in vector mosquitoes. March 2024 - NMSU Research Week.

Volunteer Lab Assistant - Environmental Biology lab
New Mexico State University

November 2022 - Jan 2023
PI: Geoffrey Smith PhD

- Conducted mass spectrometry for hydrocarbon analysis and applied fluorescence microscopy to study hydrocarbonoclastic bacteria from produced water samples, supporting postdoctoral research projects.

Rodent Lab Assistant - Cui Lab
University of Texas of the Permian Basin

Jan 2021 - May 2021
Odessa, TX

- Monitored and performed husbandry tasks for rodents under different neurological conditions such as AD, Parkinson's, MS, ASD, Epilepsy.
- Performed PCR, autoclaving and lab maintenance tasks, along with collecting phenotypical data of rodents to assist with research projects.

ACADEMIC SPECIALIZATIONS

Bioinformatics: R in statistics, R & Rshiny in bioinformatics, ML and data science in bioinformatics, systems biology(genome scale modelling and stochastic kinetics of biochemical pathways), Database management (SQL), NGS pipeline management, applied genetic analysis using public datasets (GWAS, EWAS, meta-analysis).

Microbiology/Chemistry: Organic and Biochemistry, Immunology, Medical microbiology (trained with BSL-2 organisms), Applied environmental biology, physiology & molecular biology of microorganisms.

ADDITIONAL EXPERIENCE

Teaching Assistant - College Algebra
University of Texas of the Permian Basin

Aug 2021 - Dec 2021
Odessa, TX

- Collaborated with the professor on preparation and presentation of lectures.
- Graded student assignments and held office hours to provide constructive feedback.

Athlete (Tennis)

- Competed in the international tour and in NCAA Division I Men's Tennis, earning *Junior of the Year (2023)* nomination, recognizing performance and sportsmanship.
- Served as team captain during senior year, mentoring teammates to foster collaboration and performance excellence.
- Community service of 120 hours overall which included work in elementary schools and hospitals.
- Coached and mentored 50+ kids and adults in tennis, individually and via camps, currently co-mentoring juniors in the tour.

HONORS AND AWARDS

Dean's Honor Roll	<i>University of Texas of the Permian Basin (2020 – 2021)</i>
LSC Commissioner's Honor Roll	<i>University of Texas of the Permian Basin (2021)</i>
International Competitive Scholarship Recipient	<i>New Mexico State University (2021 – 2024)</i>
C-USA Commissioner's Honor Roll	<i>New Mexico State University (2023 – 2024)</i>
C-USA Jim Castaneda Postgraduate scholarship (Nomination)	<i>New Mexico State University (2023 – 2024)</i>
Bioinformatics MS Merit Scholarship Recipient	<i>Boston University (2024-25)</i>